

Cantor's Infinities

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1 Introduction

Suppose you have two sets of numbers: one with 5 numbers and the other with 10 numbers. Which set is larger? Obviously, the set with 10 numbers is larger. What if one set had 20 numbers and the other had 21? Well, once again, the set with 21 numbers is larger. How about the set of all natural numbers and the set of all real numbers? Well, now it is trickier. This question was so tricky in fact, that only 128 years ago was this question even taken into serious consideration and solved.

The concept of infinities terrified mathematicians for thousands of years. The ancient Greeks were known to fear infinities. One of Pythagoras' friends was thrown off of a boat for proving that $\sqrt{2}$ was an irrational number whose decimal representation would go on forever without a pattern, while Zeno's paradoxes attempted to describe infinitude as an unknown that would create more questions than answers. Even in the late 19th century, the father of infinities, Georg Cantor, fearing for his reputation, had to write his ideas and theorems about infinities in the appendix of his paper about number theory.

2 Definition

In order to deal with infinities, some definitions are needed. A set is a collection of unique elements that are grouped together (e.g. set $X = \{1, 2, 4, 8, 16\}$, set $Y = \{a, b, c, d, e, f, g, h\}$, set $Z = \{this, is, a, set\}$) which is unordered. An injection from set A to set B is when every element in set A can be assigned to a unique element in set B . A surjection from A to B is when every element in B can be assigned to a unique element in A (so a surjection from A to B is the same as an injection from B to A). A bijection is when every element in A can be mapped to a unique element in B such that there are no elements left over in B .

We also have to get comfortable with sets that have infinite elements. For example, any set X where $|X| = |\mathbb{N}|$ will be defined as a set with countably infinite elements (the absolute value symbol counts the number of elements in a set, or the cardinality of a set). We say that two sets, even if they are infinite, have the same cardinality if there exists a bijection between them. For example, $|\mathbb{N}| = |\text{even positive numbers}|$, since we can map every element in $\mathbb{N}$ to every even number through the function $f(x) = 2x$.

As we are finding out, even if an infinite set contains another infinite set, they can still have the same cardinality. For example, the set of integers is countably infinite. We can assign 1 to 0, 2 to 1, 3 to -1 , 4 to 2, 5 to -2 , and so on, such that $f(n) = (-1)^2 \lfloor n/2 \rfloor$. We have a bijection, since every element in $\mathbb{N}$ can be assigned to every unique element in $\mathbb{Z}$. We can use this similar method

of snaking back and forth to prove that the set of rational numbers is countable by snaking from the top left corner of this infinite grid:

Figure 1: An easy proof that rational numbers are countable.

	1	2	3	4	5	6	7	8	...
1	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	$\frac{1}{8}$	...
2	$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\frac{2}{6}$	$\frac{2}{7}$	$\frac{2}{8}$	...
3	$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{3}{6}$	$\frac{3}{7}$	$\frac{3}{8}$	...
4	$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\frac{4}{6}$	$\frac{4}{7}$	$\frac{4}{8}$	...
5	$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\frac{5}{6}$	$\frac{5}{7}$	$\frac{5}{8}$	...
6	$\frac{6}{1}$	$\frac{6}{2}$	$\frac{6}{3}$	$\frac{6}{4}$	$\frac{6}{5}$	$\frac{6}{6}$	$\frac{6}{7}$	$\frac{6}{8}$	...
7	$\frac{7}{1}$	$\frac{7}{2}$	$\frac{7}{3}$	$\frac{7}{4}$	$\frac{7}{5}$	$\frac{7}{6}$	$\frac{7}{7}$	$\frac{7}{8}$	...
8	$\frac{8}{1}$	$\frac{8}{2}$	$\frac{8}{3}$	$\frac{8}{4}$	$\frac{8}{5}$	$\frac{8}{6}$	$\frac{8}{7}$	$\frac{8}{8}$	...
⋮	⋮								

We can assign 1 to $\frac{1}{1}$, 2 to $\frac{2}{1}$, 3 to $\frac{1}{2}$, 4 to $\frac{3}{1}$, 5 to $\frac{2}{2}$, and so on. We can also prove that if you have a countably infinite number of baskets with a countably infinite number of apples in each basket, then the amount of apples is countably infinite. Suppose you label each basket 0, 1, 2, 3, and so on. In each basket, label the apples as 1, 2, 3, and so on. For each natural number, break it up into digits of even powers of ten and digits of odd powers of ten (for example, 12345 would be 135 and 24). The digits in odd powers of ten would describe the basket and the digits in even powers of ten would describe the apples. So the 27th apples in the 84th basket would be described by the number 8247 (if the numbered basket has less digits than the place it holds in the basket or vice versa, you can just fill in the rest of the even or odd digits with 0's). We have our bijection.

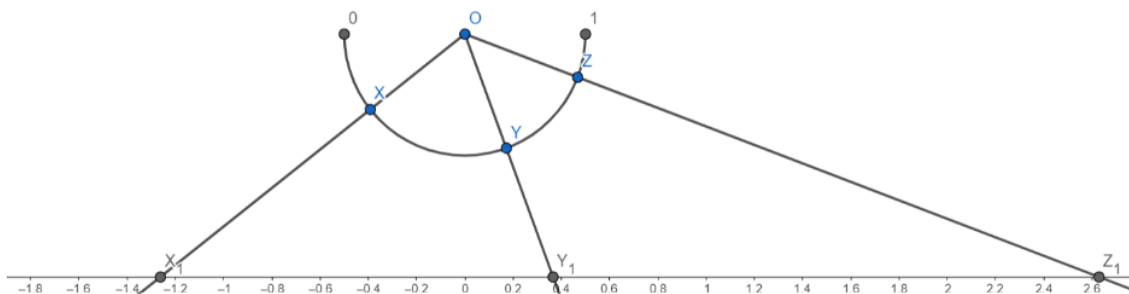
3 The Union Principle

From this last bijection, we have the Union Principle, which states that if $|A| = |B| = |\mathbb{N}|$, then $|A \cup B| = |\mathbb{N}|$. The Union Principle can be repeated multiple times, so this can be expanded to the following:

$$|A_1 \cup A_2 \cup A_3 \dots A_i| = |\mathbb{N}|$$

if $|A_i| = |\mathbb{N}|$ for any positive integer i .

Before we figure out the cardinality of the real numbers, for simplification, we will prove that the cardinality of the real numbers from 0 to 1 is equal to the cardinality of all real numbers. This can be accomplished through a simple, elegant image:



The semicircle is the segment from 0 to 1, while the line at the bottom represents all real numbers. As we can see, every real number on the semicircle can be mapped to every real number on the number line by drawing a line from the center of the semicircle to the semicircle, and then extending it to the number line.

4 Cardinality of Real Numbers

We can now find the cardinality of the real numbers. Since the cardinality of the real numbers and the real numbers from 0 and 1 are the same, we can compare the natural, countable numbers, to the real numbers from 0 to 1. We will conduct a proof by contradiction. Assume that it is possible to assign every natural number to any real number. We will write out a random list of real numbers, but for this argument, any list would work.

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1  0.283401982491...
2  0.230498123809...
3  0.912358934576...
4  0.492750740595...
5  0.933850384947...
...
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Now, if we can generate a new real number between 0 and 1 that is different from every number on this list, then we have proved that there exists a higher cardinality than the countables. If you take the first digit after the decimal point of the first number and change it to a different digit, the second digit of the second number and change it to a different digit, the third digit of the third number and change it to a different digit, and so on, we would have generated a new number which is different from the 1st (since it would have a different first digit), different from the 2nd, 3rd, and every number already on the list. Thus, we have proved that the cardinality of the reals is greater than the cardinality of the natural numbers.

But wait, a skeptical reader like yourself might wonder, some real numbers have two decimal representations. For example, 0.5 can also be written as 0.499... So when generating a new number, one might write the new number as 0.499..., even if 0.5 is a part of the list, and vice versa. So when generating a new number, we will have to exclude the digits 0 and 9 when selecting the new digits. So now we have proved that the cardinality of the reals is greater than the cardinality of the natural numbers. We will call this new cardinality uncountable infinity.

By the Union Principle, since all real numbers are either rational or irrational, and rational numbers are countably infinite, if the irrationals were countably infinite, then the reals would also

be countably infinite. And since we just proved that the reals are uncountable, then the irrationals are also uncountable. But we are not done. In fact, we have found just one out of an infinite number of new cardinalities of infinities.

5 Power Sets

A power set is the set of all subsets of a set. For example, the power set of the first 3 perfect squares starting from 1 is $\{\{\}, \{1\}, \{4\}, \{9\}, \{1, 4\}, \{1, 9\}, \{4, 9\}, \{1, 4, 9\}\}$. As we can see, there are $2^3 = 8$ elements. We can say that the power set of a set X is $P(X)$. We can prove that the size of the power set of a set with n elements is 2^n through induction:

Base Case: $n = 1$

$$|\{\{\}, \{1\}\}| = 2$$

Assume that the size of the power set of a set with $n = k$ elements is 2^n

Induction step: Prove that this holds for $n = k + 1$

Assume that the set is the natural numbers from 1 to $k + 1$. Each subset will either have or not have $k + 1$. So the size of this power set is the size of the power set of $n = k$ plus the size of the power set of the power set of $n = k$ but each subset has $k + 1$. Since each set has 2^k elements, the size of the power set of the natural numbers from 1 to $k + 1$ is $2^{(k+1)}$.

Now, let's conduct a similar proof to that which we did before with the reals. We can assign each element in an infinite set (we will use the naturals as an example) to a subset of that very set. Such a subset will be represented as an infinitely long binary number, where a 0 signifies an element not being present in the subset and a 1 signifying the presence of that element in a set. So for example, if we have the set $\{0, 1, 2, 3\}$, the subset $\{0, 2, 3\}$ would be represented as 1011. We can create such a chart:

1	1000101110111...
2	0110100100010...
3	0100011110001...
4	1010001110100...
5	1001100011011...
...	

We can do the same diagonal trick as with the reals, where we flip the 1st number of the subset assigned to 1 and flip it (a flip would change a 0 to a 1 and a 1 to a 0), flip the 2nd number of the subset assigned to 2, and so on. We have now generated a new subset that is unique from the rest. This means that the power set of a set has a higher cardinality than the set itself.

This can generate infinitely many infinite cardinalities, ranging from $\mathbb{N}$ to $P(\mathbb{N})$, which is just uncountable infinity, to $P(P(\mathbb{N}))$, to $P(P(P(\mathbb{N})))$, and so on.

The notation we will use to describe the sizes of infinities is the Hebrew letter ($\aleph$). Countable infinity is written as $\aleph_0$, the first uncountable infinity is written as $\aleph_1$, and the rest follows until

infinity.

6 Problems

1. Prove that if A is an infinitely large set, then $|A| \geq |\mathbb{N}|$.
2. Prove that there is a bijection between the number line and the coordinate plane (hint: don't forget to watch out for the fact that there are two decimal representations for some numbers, such as 0.5 being equal to $0.499\dots$).
3. Prove that the cardinality of the reals is the same as the cardinality of complex numbers (Hint: use the result from problem 2).
4. Prove that there if there is both a injection and surjection from A to B , then $|A| = |B|$. (Schröder-Bernstein Theorem). Very difficult!

References

“An Easy Proof That Rational Numbers Are Countable.” Home School Math, n.d.
<https://www.homeschoolmath.net/teaching/rational-numbers-countable.php>.